COURSE INFORMATION

School/Faculty:	**Applied Computing & AI/Faculty Computing**		
Program name:	**Bachelor of Computer Science (Data Engineering) with Honours**		
Course code:	**SECP2613**	**Academic Session/Semester:**	**20252026/2**
Course name:	**System Analysis & Design (WBL)**	**Pre/co requisite (*course name and code, if applicable*):**	**-**
Credit hours:	**3**		

Course synopsis	**This course introduces** students to systems analysis and design skills. **Students will explore** structured methodology, including planning, analysis, and design phases. **The course also provides** industry exposure to solve real-life case studies. **The course is delivered through** lectures, active learning, and work-based learning. **By the end of the course**, students will be able to design a prototype, including form input, output and interface design based on the planning and analysis output.			
Course coordinator (*If applicable*)	Dr. Muhammad Iqbal Tariq Idris			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail** *@utm.my*
	Dr. Muhammad Iqbal Tariq Idris	05-20 (N28a)	013-222 4260	miqbaltariq
	Dr. Aryati Bakri	402-10 (N28)	011-5927 7750	aryati

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	* CLO	PLO ** (MQF Cluster Code)	*** Taxonomies and **** Graduate Attributes	T&L methods	*****Assessment methods
CLO1	Describe the fundamental system analysis and design (SAD) principals in SAD system lifecycle through system thinking in solving for real-life case studies.	PLO1 (C1)	C2 SC1	Lectures	T, F, Asg
CLO2	Evaluate strategic thinking solutions for case study problems by applying SAD principles using structured methodology.	PLO2 (C2)	C6 SC2	Lectures, Work-based learning	T, F, Asg
CLO3	Construct the high-fidelity prototype using a collaborative interface design tool using through integrated problem solving.	PLO3 (C3A)	P3 SC3	Lectures, Work-based learning	PR, Ind
CLO4	Organizing the SAD principles in a collaborative setting for a case study solution using learned designing techniques and design tools.	PLO8 (C3F)	A4 A(CC2)	Work-based learning	S&P

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
* Up to 5 CLO
Refer *** Taxonomies of Learning and **** UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement
***** Asg – Assignment; T – Test; Ind-Industry; PR – Project; F – Final Exam; S&P – Self & Peer Review.

** **MQF Cluster Code**
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	In-class activities such as Think-Pair-Share, Jigsaw, Gallery Walk, Troika Consulting, The Journalistic Six, and Bad Idea Brainstorm are used to promote engagement, collaboration, and deeper understanding of course concepts.
2	Work-based learning	Conducted in a group project assignment. Students, in a group of 3 or 4, are required to design a high-fidelity prototype for a selected company. Students are required to go through the structured methodology starting from determining the system requirements until design the system. Students shall apply all relevant knowledge acquired throughout the semester on the project. Students are assessed on their ability to deliver a creative system design solution that will be able fulfill the requirements identified.

Transferable skills (*generic skills learning in course of study which can be useful and utilised in other settings*) :

Able to share, influence and guide other members of team.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	**TOPIC I SYSTEMS ANALYSIS FUNDAMENTAL** **Part 1** - Organizational impact on Information System - Types of Information Systems **Part 2** - Systems analyst role **Part 3** - Types, trend and approach towards developing information system	CLO1					3				1	2	6
Week 2	**TOPIC II PROJECT PLANNING PROCESS** **Part 1** - Project feasibility studies - Project initiation **Part 2** - Project planning and control - Project schedules and techniques (Gantt Chart, PERT Chart, Critical Path Method) - WBS **Part 3** - Project cost management (CBA) - Project Management - Introduction to Project Management Software Assignment 1: 2%	CLO1, CLO2	2		1						1	2	6

Week 3	**TOPIC III INFORMATION REQUIREMENT TASK** **Part 1** - Information Gathering – Interactive Methods **Part 2** - Information Gathering – Unobtrusive Methods	CLO1, CLO2, CLO3	2		1						1	2	6
Week 4	**Industry Day 1** (Real-life case studies sharing & P1 briefing)	CLO2	2		1						1	2	6
Week 5	**TOPIC IV THE ANALYSIS PROCESS** **Part 1** - Introduction to Data Flow Diagrams (DFD) Assignment 2: 2% **Part 2** - DFD Errors (P1 submission: 10%, P2 briefing)	CLO1, CLO2	2		1						1	2	6
Week 6	**Part 2 (Continue)** - Introduction to DFD Enterprise Architect Mid Term Test: 15%	CLO1, CLO2	2		1						1	2	6
Week 7	**Part 3 (Continue)** - From Logical to Physical DFD Partitioning DFD Assignment 3: 4%	CLO1, CLO2	2		1	2					1	2	8
Week 8	**MID SEMESTER BREAK**												
Week 9	**Part 4** - Describing Process Specification and techniques (Decision trees, Decision table, Structured English) - Design Structure Chart (P2 submission: 10%, P3 briefing)	CLO1, CLO2, CLO3	2		1	1					2	2	8
Week 10	**TOPIC V THE DESIGN TASK** **Part 1** - Design System Architecture - Designing User Interfaces - Storytelling HCI - Designing Effective Output - Designing Effective Input	CLO1, CLO3	2		1	2					2	2	9
Week 11	**Part 2 Continue** - Introduction of designing tools (Figma)	CLO1, CLO3	2		1	2					2	2	9
Week 12	**TOPIC VI SYSTEMS IMPLEMENTATION** **Part 1** - User Testing and acceptance test - Training plan & strategies	CLO2, CLO3	2		1	2					2	2	9
Week 13	**TOPIC VI SYSTEMS IMPLEMENTATION (Continue)** **Part 2** - Implementation plan & strategies	CLO1, CLO3	2		1	2					2	2	9
Week 14	**Project assessment and group presentation** (P3 submission: 15%, Self & Peer Review: 5%)	CLO3, CLO4	2		1	2					2	2	9

Week 15	**Industry Day 2** - Prototype demonstration (Industry: 7%)	CLO3	2		1	2					1	2	8
												SUB-TOTAL SLT :	**105**

	Continuous Assessment	**%**	**Face-to-Face (F2F)**		**Non F2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online** (Synchronous)	**Online** (Asynchronous)	**Others**	
1	Assignment	8	3		1	2	6
2	Self & Peer Review	5	5		2	2	9
3	Project	35			4	6	10
4	Test	15	3		1	2	6
5	Industry	7	3		1	2	6
						SUB-TOTAL SLT :	**37**

	Final Assessment	**%**	**Face-to-Face (F2F)**		**Non F2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online** (Synchronous)	**Online** (Asynchronous)	**Others**	
1	Final Exam	30	3			5	8
						SUB-TOTAL SLT :	**8**
						SLT for Assessment :	**45**
						GRAND TOTAL SLT :	**150**

A	% SLT for F2F Physical Component :	47.33
B	% SLT for Online & Independent Learning Component :	52.67
C	% SLT for Online Component :	21.33
D	% SLT for All Practical Component :	8.67
D1	% SLT for F2F Physical Practical Component :	8.67
D2	% SLT for F2F Online Practical Component :	0.00
Please tick (✓) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (*e.g., software, nursery, computer lab, simulation room etc*)

Computer lab that is equipped with analysis software (Enterprise Architecture) and design software (Figma).

References (*include required and further readings, and should be the most current*)

Main reference:
Kendall & Kendall. (2023). System Analysis & Design, 11th edition, Essex: Pearson Education Limited.

Additional references:
Valacich, George & Hoffer (2021). Essentials of Systems Analysis & design, 6th Edition, Essex: Pearson Education Limited.
Tilley & Rosenblatt (2017). System Analysis and Design. 11th Edition, Singapore. Cengage Technology Edition.Online

Online:
UTM e-learning (http://elearning.utm.my)

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (*if applicable*)

All students must adhere to the following to pass a computing course:

i. Pass both Continuous Assessment AND Final Assessment.
ii. Get at least minimum pass marks following UTM regulation (40% or D+).

The passing mark for:

- Continuous Assessment is 30% of continuous marks.
- Final Assessment is 20% of the final marks.

If a student does not achieve the passing mark for Continuous Assessment OR the passing mark for the Final Assessment, then the student's overall final mark will be set below the UTM passing mark.

Please refer to UTM Academic Handbook and Guidebook for other rules and regulations.

Disclaimer:

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While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by :			**Certified by:**		
Name	:	Dr. Muhammad Iqbal Tariq Idris	**Name**	:	Dr. Sharin Hazlin Huspi
Signature	:		**Signature**	:	
Date	:	12 Mar 2026	**Date**	:	